



Evaluating Temporal Dimensionality Reduction Methods for ERP-Structured EEG Data

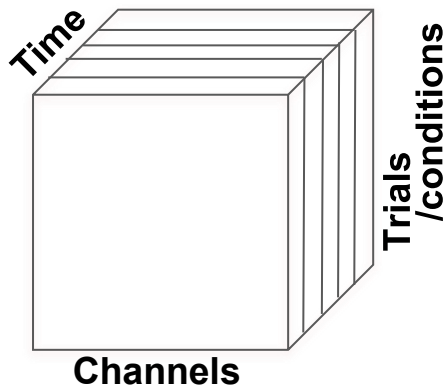
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Visualization Institute of the University of Stuttgart (VISUS)

Motivation

Why is EEG analysis difficult?

High-dimensional data:



✗ Traditional plots can not show full structure at a time (time, spatial structure, or condition)

✗ Patterns missed across dimensions

✗ Manual comparison doesn't scale with complex designs

How to analyze?

**Dimensionality Reduction:
Compress all dimensions into two-dimensional space**

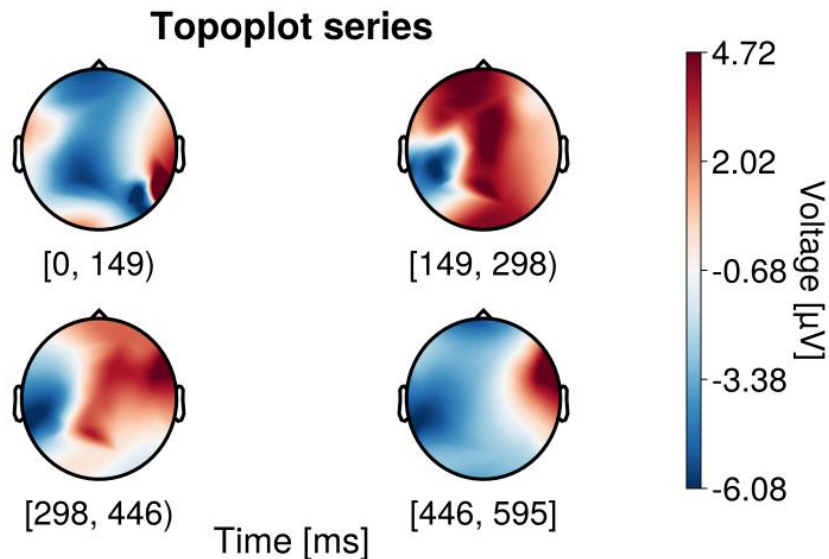
- ✓ Enable exploratory analysis
- ✓ Uncovers meaningful patterns in complex data
- ✓ Guide hypothesis testing

Motivation: Visualization Methods Comparison

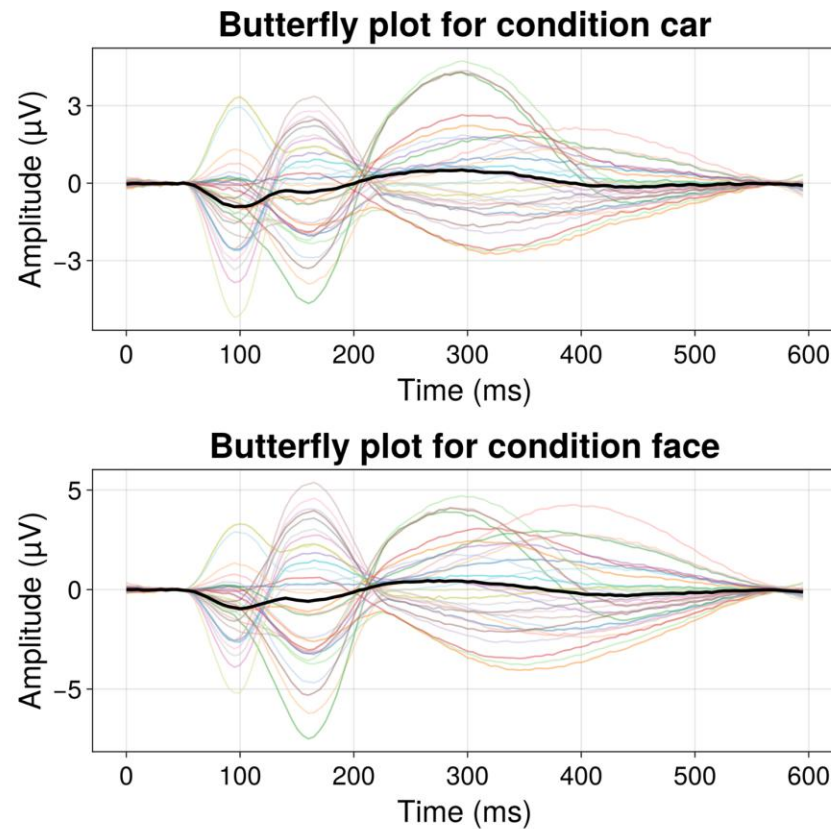
Aspect	Butterfly Plot	Topoplot Series	Dimensionality reduction
Amplitude over time	✓	✓	✓
Individual channels	✓	✓	⚠ Channels collapsed meaningfully
Spatial layout	✗	✓	✗
Multiple conditions	⚠ Clutter	⚠ Many plots	✓
Continuous time	✓	⚠ Snapshots	✓

Dimensionality reduction summarises structure across channels, conditions, trials, and time in a single representation

Standard visualizations of simulated data



Grand average analysis using topoplot series



Butterfly plots across the epoch per condition



Research Questions

RQ1. To what extent do time-aware dimensionality reduction methods (BCNE and TPHATE) better preserve temporal structure in ERP-structured EEG data compared to time-agnostic methods (PCA, tSNE, UMAP, PHATE)?

RQ2. Which metrics provide the most informative assessment of temporal preservation in dimensionality reduction?



Hypotheses



Time-aware methods will significantly outperform time-agnostic methods



Representation of the continuous ERP component trajectory (P100–N170–P300–N400) and its divergence across conditions

Data simulation using *UnfoldSim.jl*

UnfoldSim generates EEG data with known temporal structure and categorial/continuous effects

Experimental design

- 2–10 categorical conditions
- 2–10 continuous variables with levels (–5 to +5);
- 32 or 64 channels (biosemi);
- 200 Hz sampling rate;
- 0–595 ms epochs.

ERP	Latency	Driven by	Source
P100	100 ms	Intercept only	Right Occipital Pole
N170	170 ms	Categorical variable	Left Postcentral Gyrus
P300	300 ms	Continuous Variable with level	Left Superior Frontal Gyrus
N400	400 ms	Categorical variable	Right Subcallosal Cortex

Methods Overview

Methods	How it works
Time-agnostic methods	
PCA	Linear projection onto directions of maximum variance
t-SNE	Groups time points by amplitude, ignoring time order
UMAP	Nearest-neighbor graph; preserves global structure
PHATE	Diffusion-based on similar activity patterns, no time order
Time-aware methods	
T-PHATE	Similar activity patterns + temporal order (Autocorrelation)
BCNE	Temporal smoothing, channel mapping, CNN Recursive training

Recap: PHATE vs TPHATE

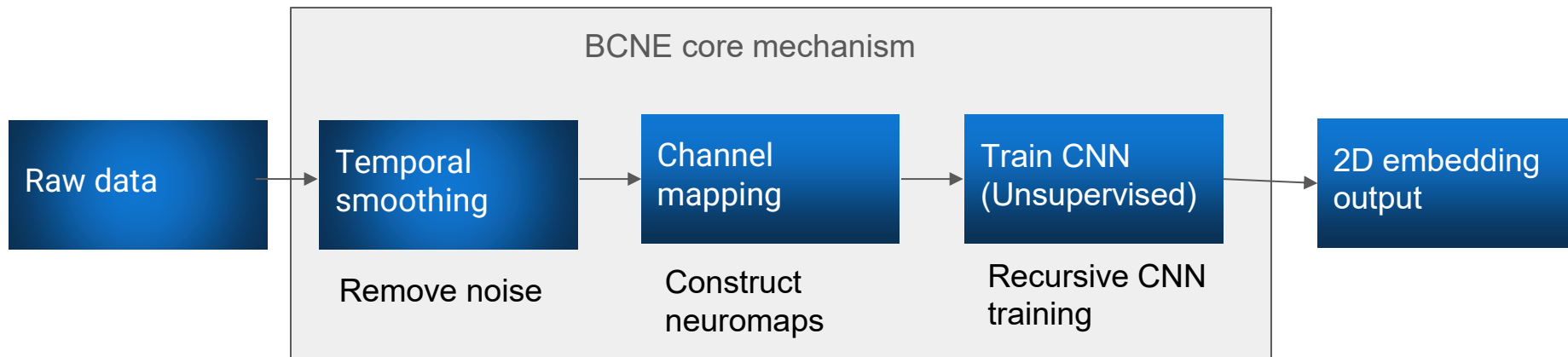
PHATE (Potential of Heat Diffusion for Affinity-based Trajectory Embedding)

- Uses diffusion geometry to capture both local and global structure in high-dimensional data
- Designed to preserve continuous trajectories and branching structures.

T-PHATE (Temporal PHATE)

- PHATE + time autocorrelation
- Time autocorrelation: how a signal is related to its past timestamp (T vs $T-1$)

Recap: Brain-dynamic Convolutional-Network-based Embedding (Unsupervised learning)



Results

Reference Dataset for results

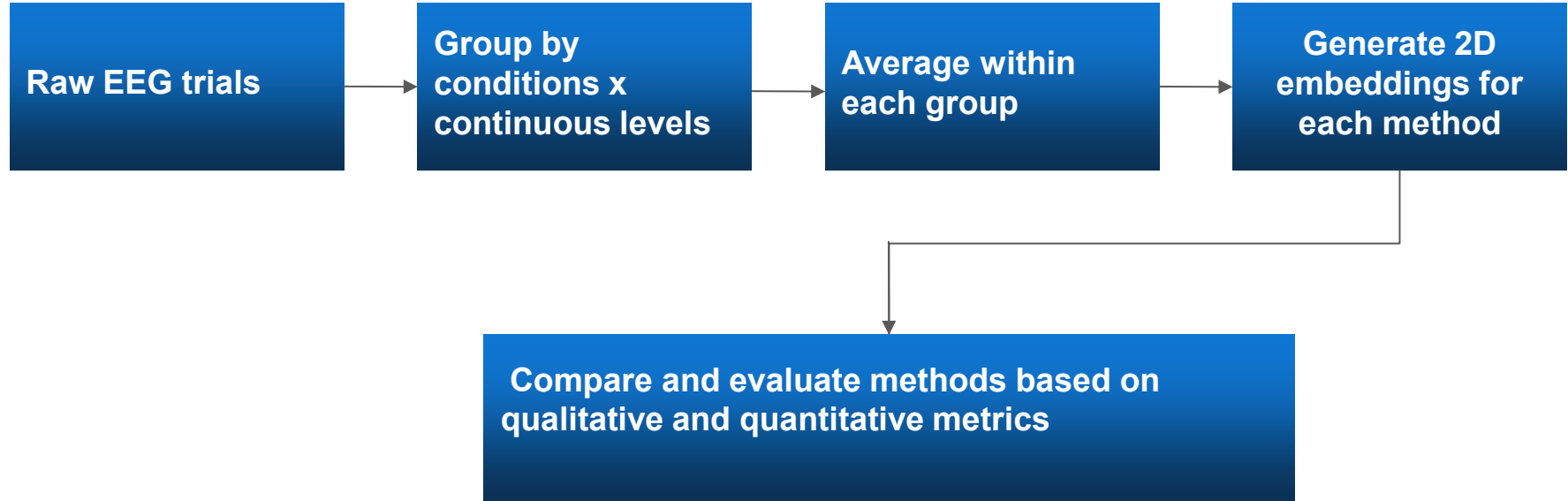
2 conditions: face and car

10 continuous variables levels: -5 to 5

4 ERP components

Row	continuous	condition
	Float64	String
1	2.22222	face
2	4.44444	car
3	3.88889	car
4	1.11111	car
5	0.555556	car

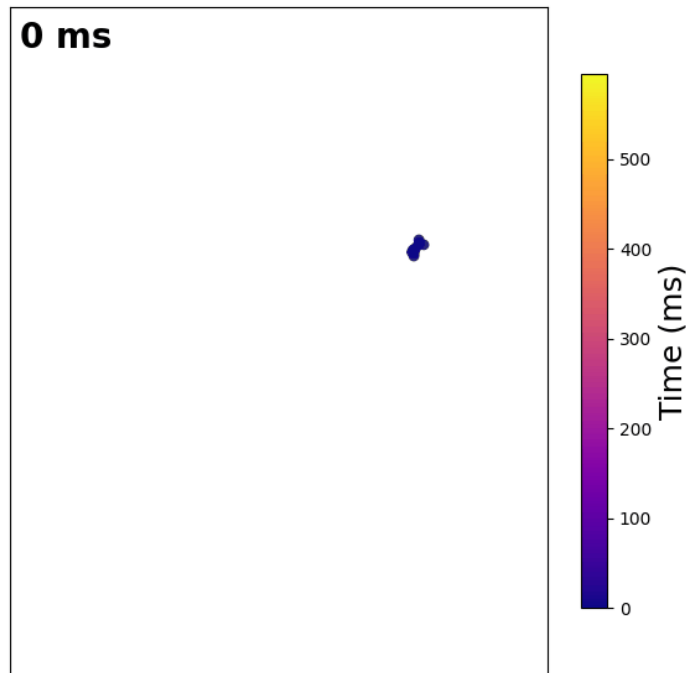
Approach 1: Average by condition



Reading the embedding

- A single dot: the brain's overall signal pattern at **one single timepoint**.
- Closed loop: the brain pattern returns close to its baseline by the end of the window.

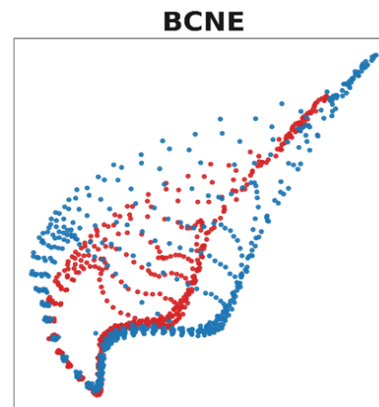
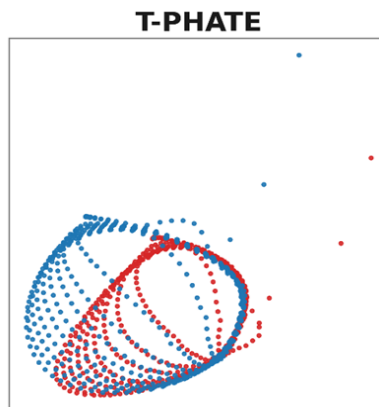
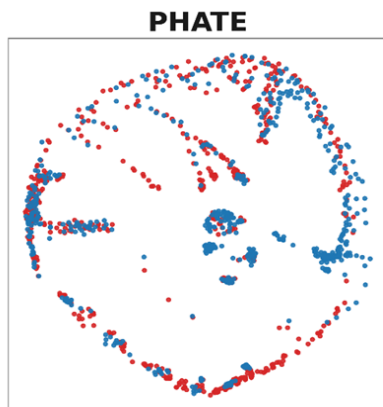
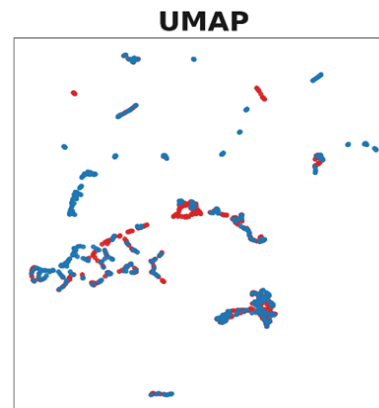
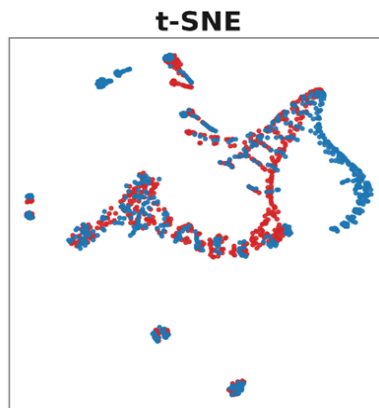
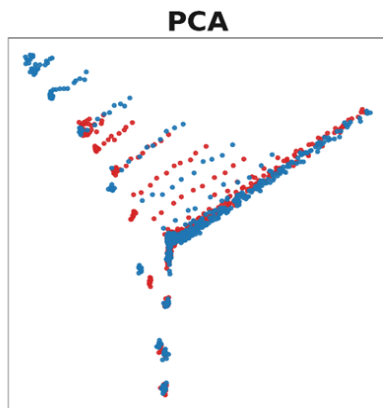
T-PHATE - Face Condition Over Time



10 different loops = 10 continuous variables.

Condition separation

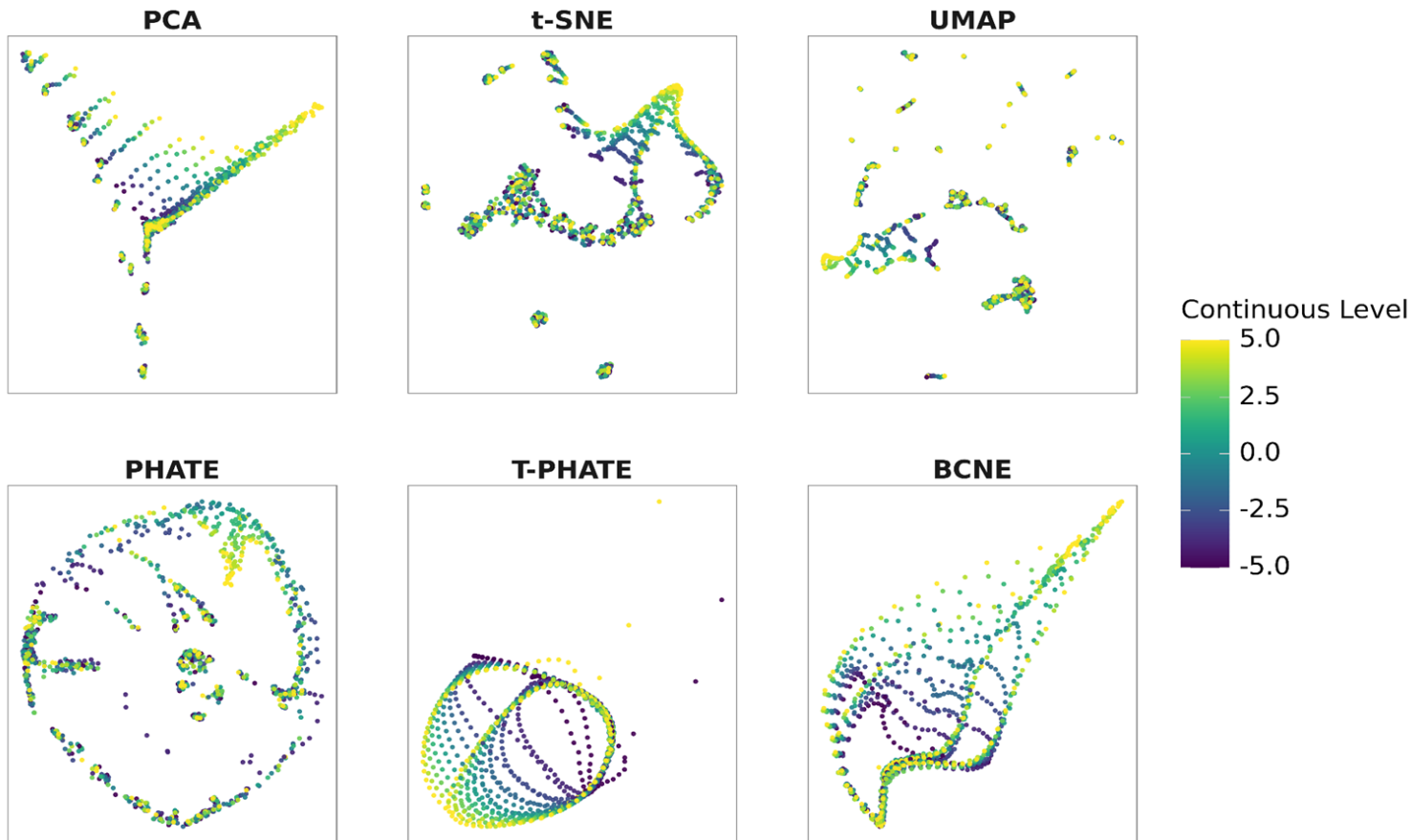
Approach 1



Condition
● Car
● Face

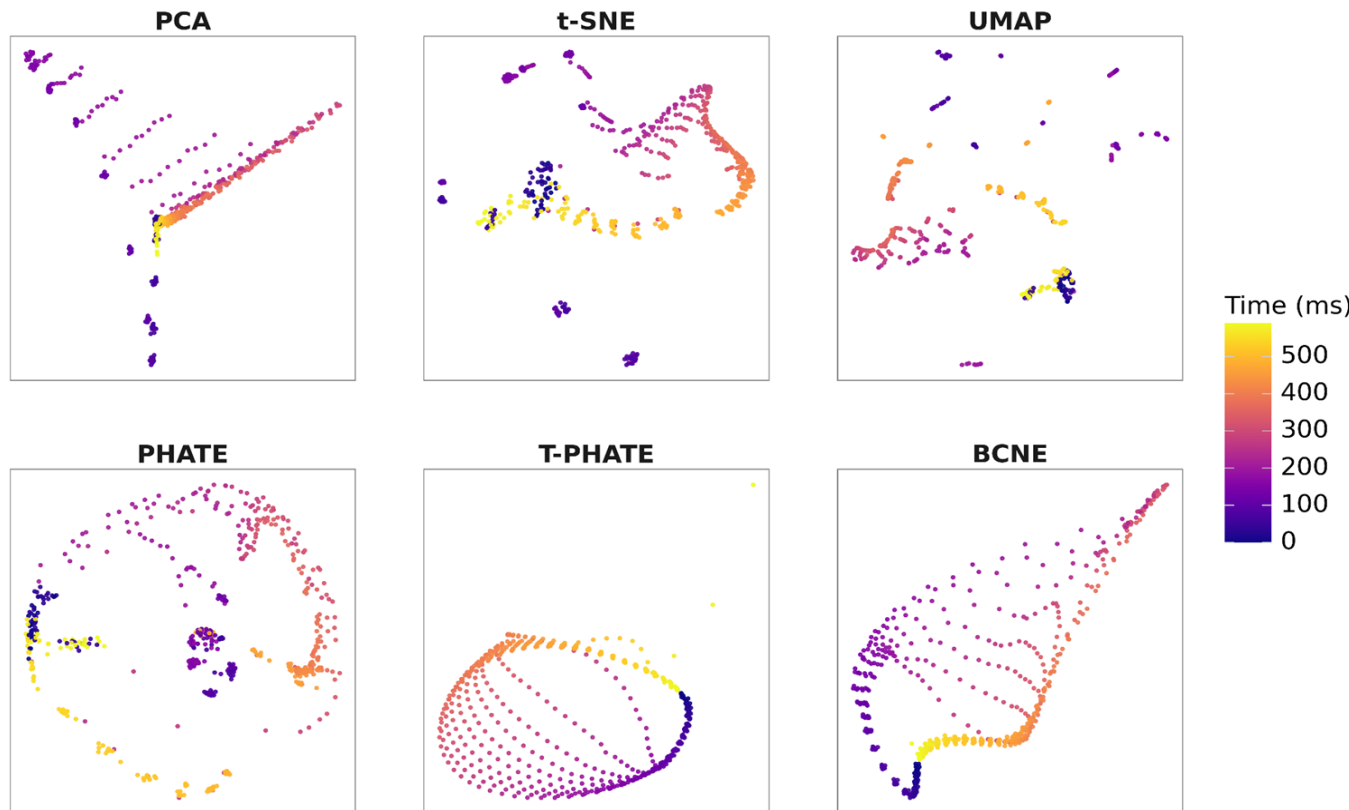
Continuous levels distinction

Approach 1



Temporal Structure Preservation for *face* condition

Approach 1

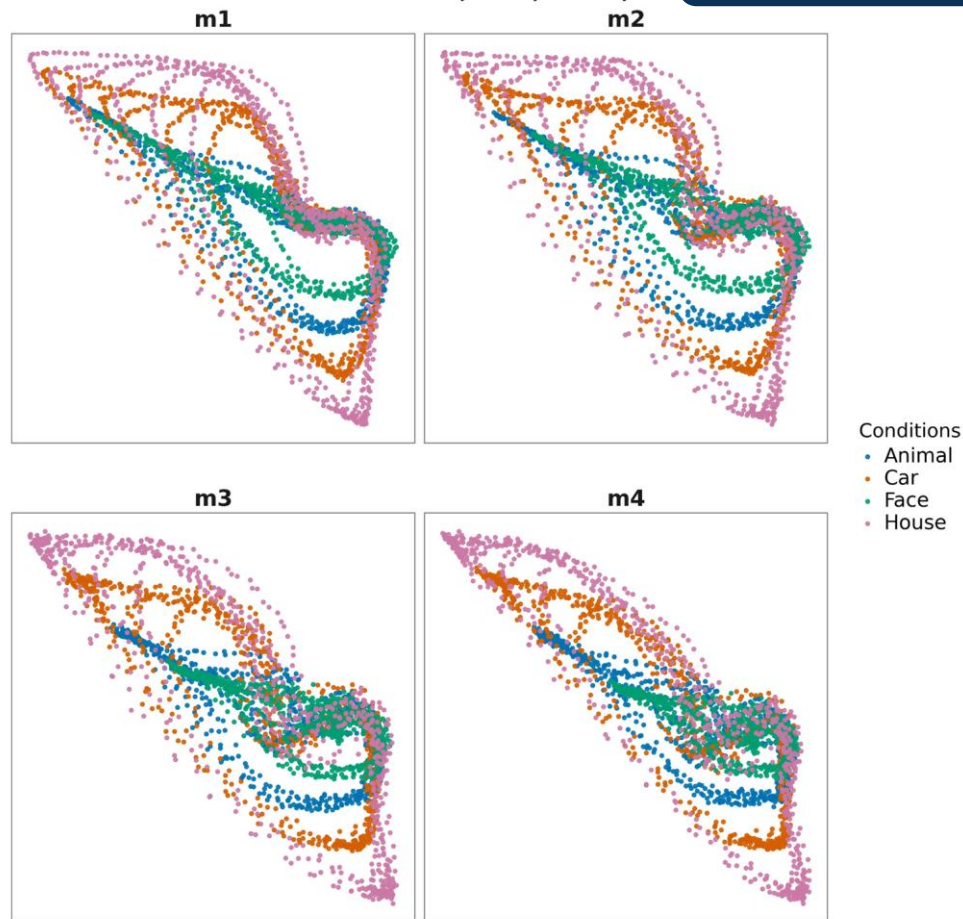


RQ1: To what extent do time-aware dimensionality reduction methods better preserve temporal structure in structured EEG data compared to time-agnostic methods?

Noise Robustness

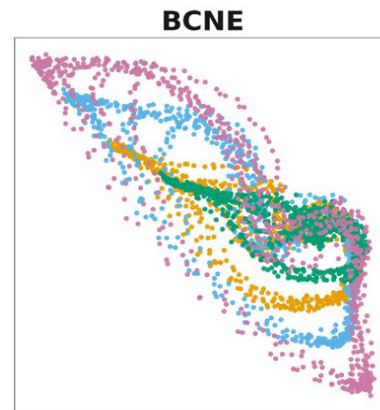
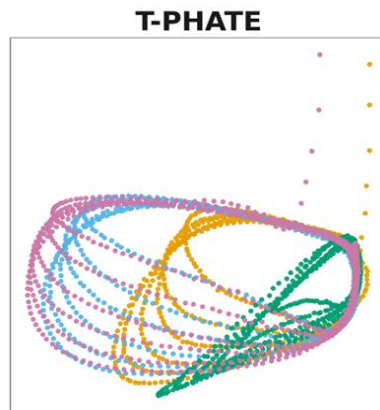
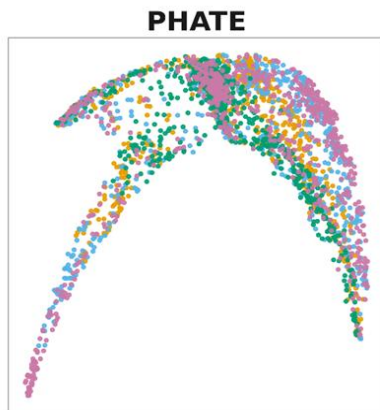
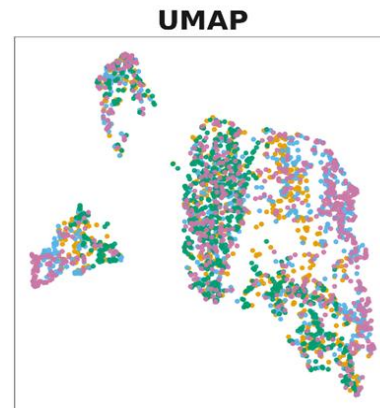
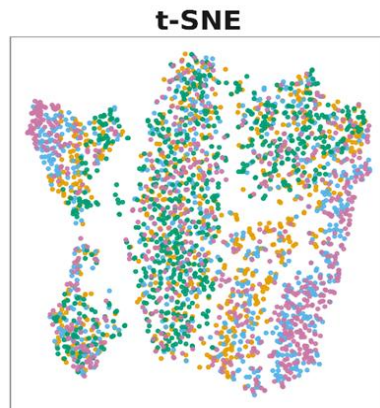
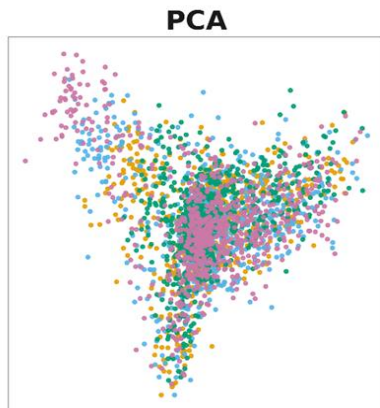
Approach 1

- The BCNE method was able to separate multiple conditions even with a higher noise level (noise level: 7)



Noise Robustness

Approach 1



Condition

- Animal
- Car
- Face
- House

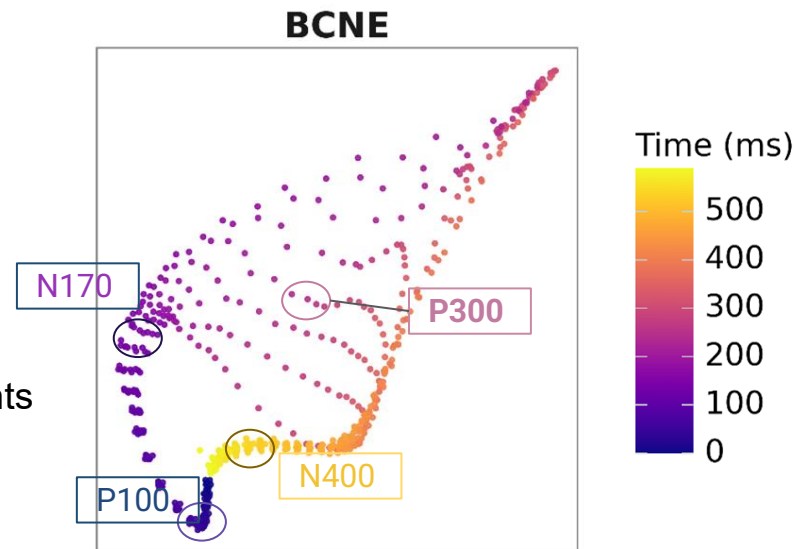
Evaluation Quantitative

	K-Nearest Neighbors	Trustworthiness	Continuity
PCA	0.294	0.973	0.991
t-SNE	0.395	0.996	0.993
UMAP	0.395	0.992	0.992
PHATE	0.374	0.986	0.993
T-PHATE	0.48	0.967	0.979
BCNE m1	0.48	0.965	0.978
BCNE m4	0.599	0.962	0.969

BCNE achieves the highest K-NN score (0.599), confirming it best preserves neighbourhood structure across all methods

Qualitative Evaluation

- ✓ **Condition Separation**
Face vs. Car distinction visible
- ✓ **Trajectory Smoothness**
- ✓ **Component Detectability**
P100, N170, P300, and N400 visible at expected time points
- ✓ **Temporal Structure**
Preservation of temporal ordering and timing

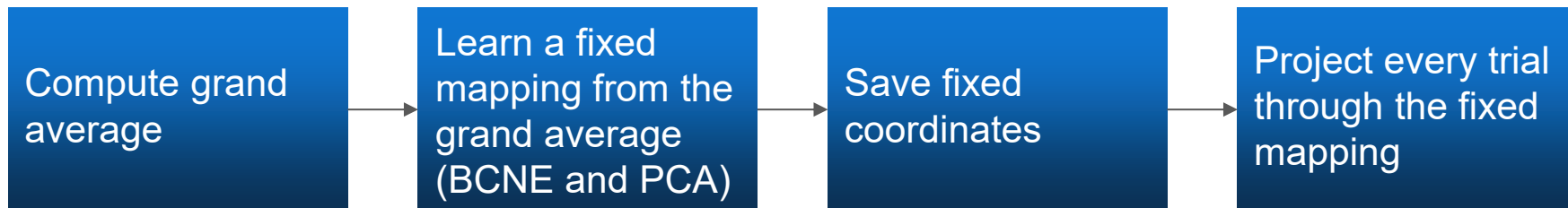


RQ2. Which metrics provide the most informative assessment of temporal preservation in dimensionality reduction?

Additional Results

Approach 2: Trial-level projection

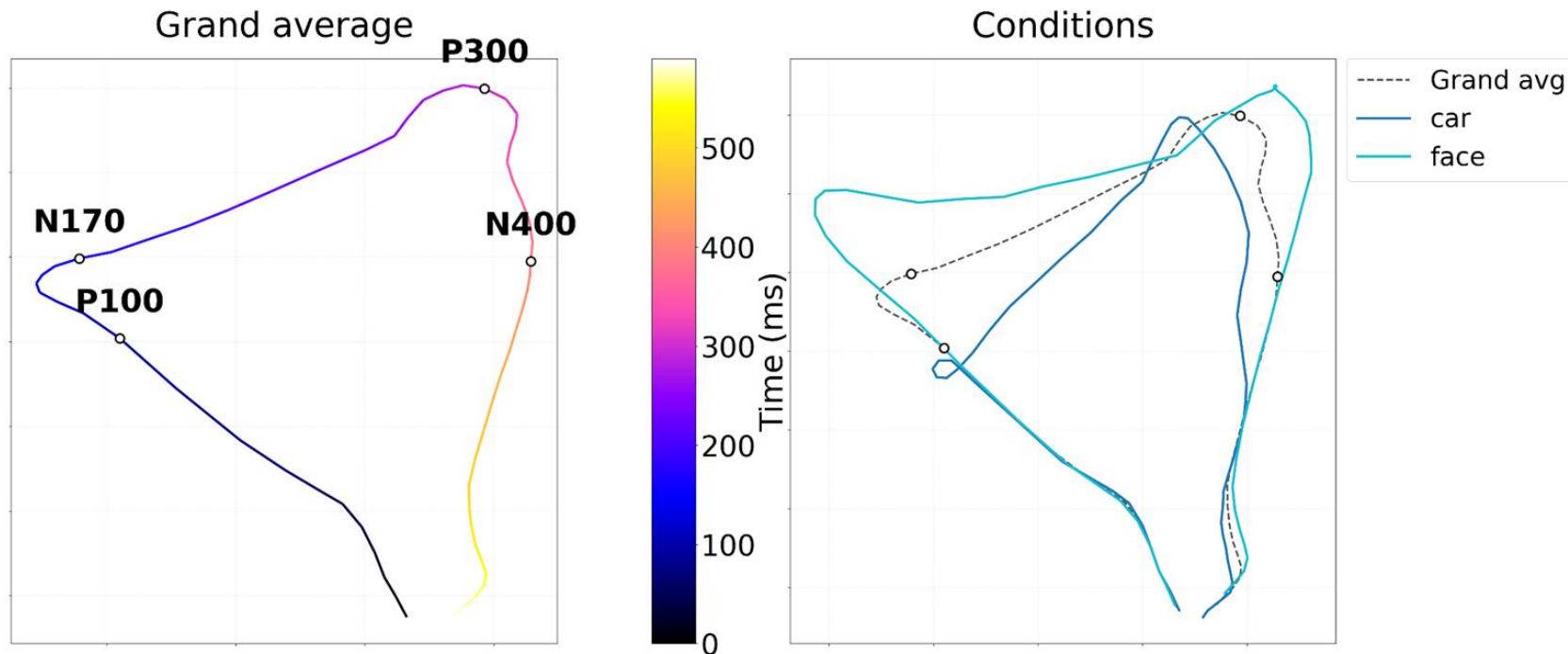
A fixed mapping is learned once from the grand average, then applied to every individual trial, preserving trial variability



Trial-level projection to PHATE, TPHATE and UMAP is a part of future scope

Train grand average and condition mean averages

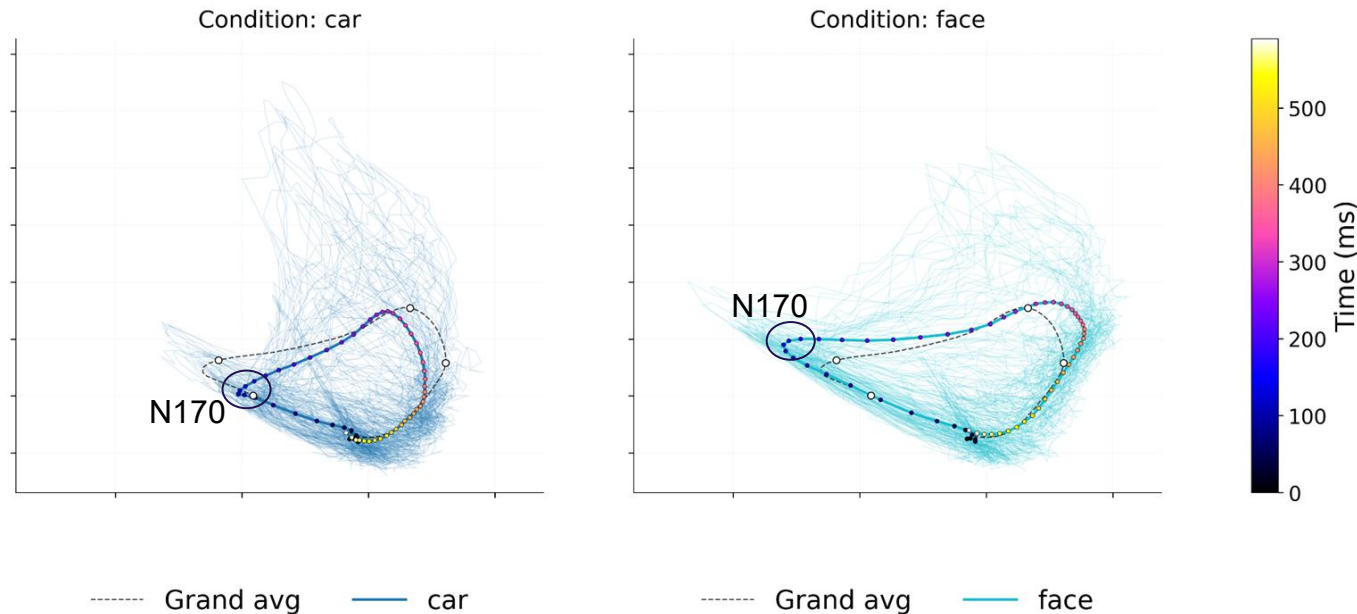
BCNE - Condition averages



Trial-projection shows condition separation

Approach 2

BCNE - Individual Trial Trajectories

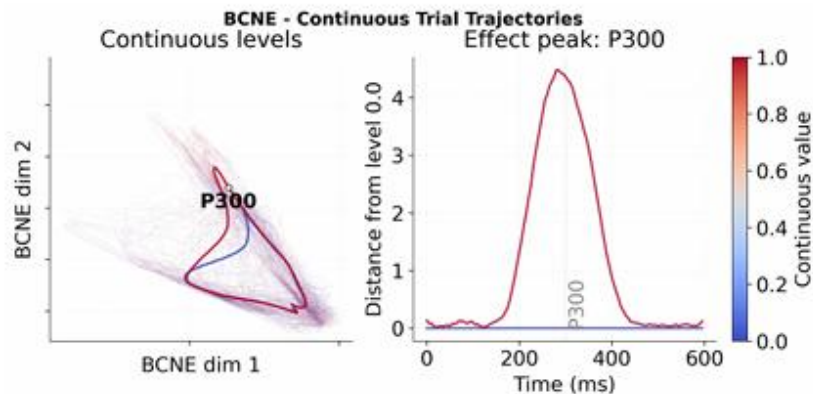
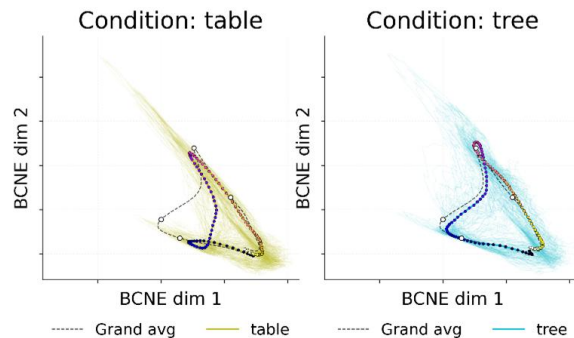
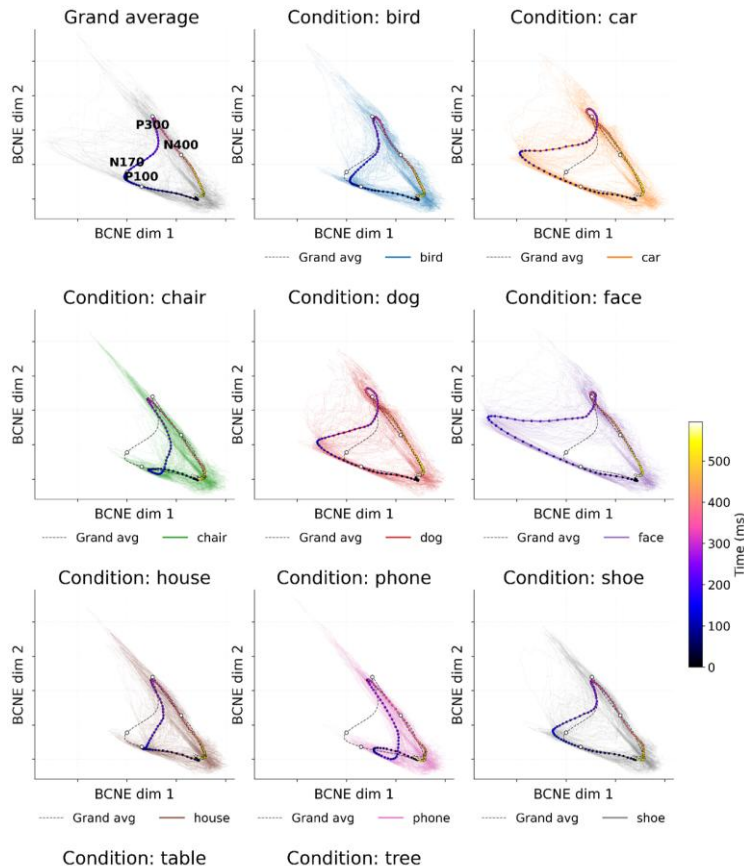


Halo transparent lines :Individual trials projected through fixed coordinates of grand avg BCNE

Bold lines: Average of halo lines

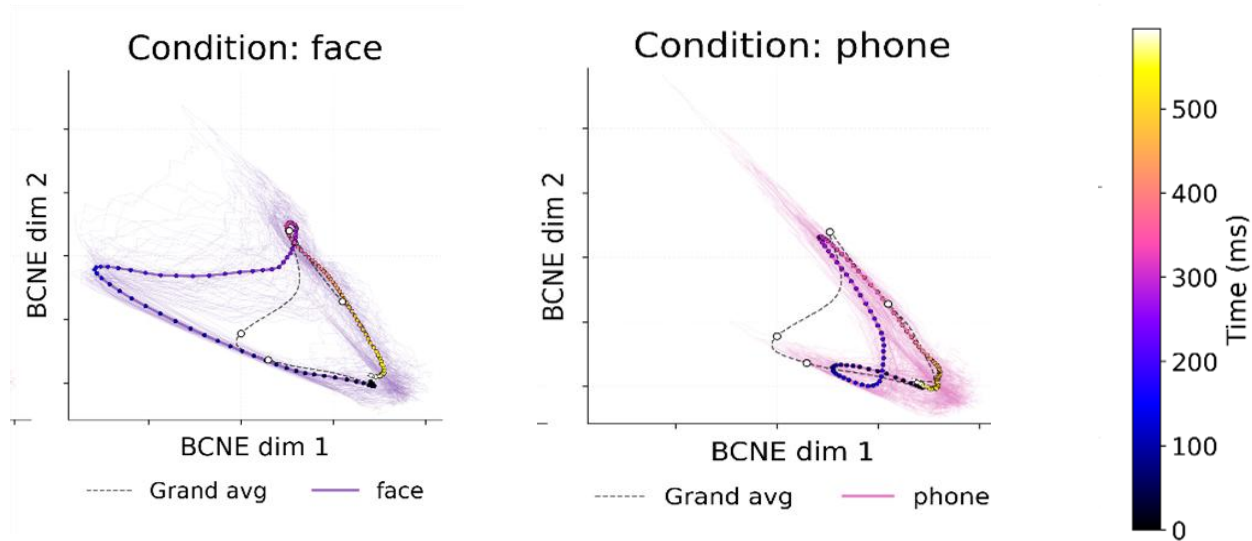
Scale to ten conditions, two continuous variables

Approach 2



Face vs Phone condition embeddings

Approach 2

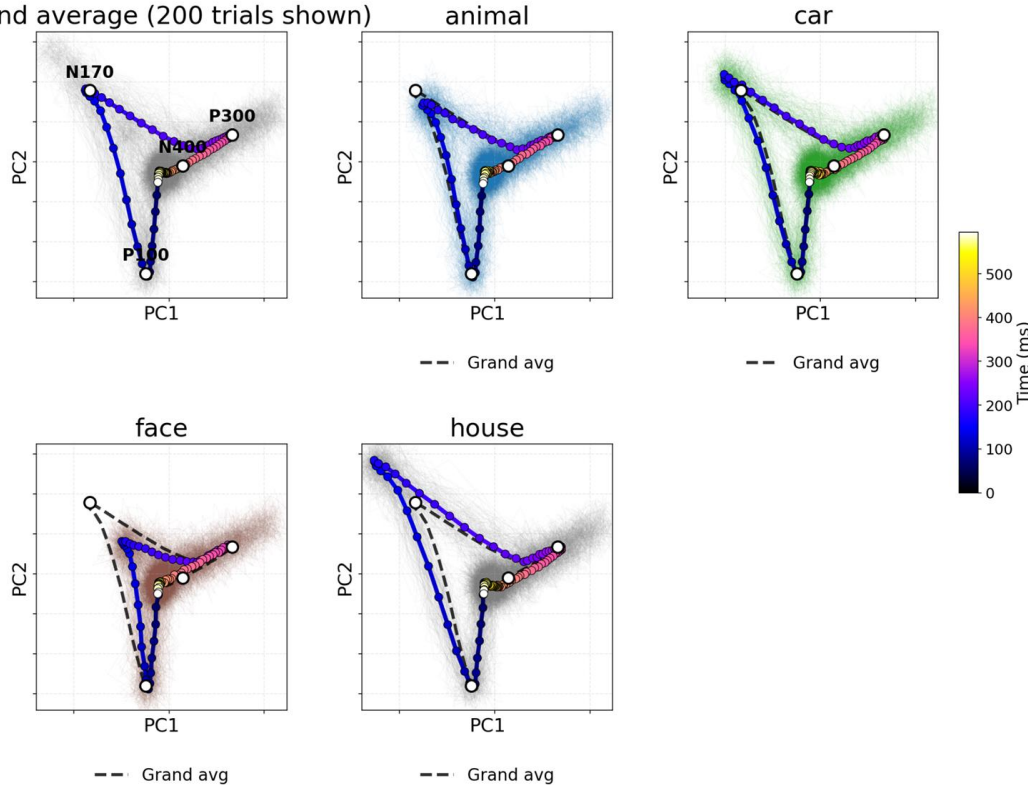


PCA baseline comparison

Approach 2

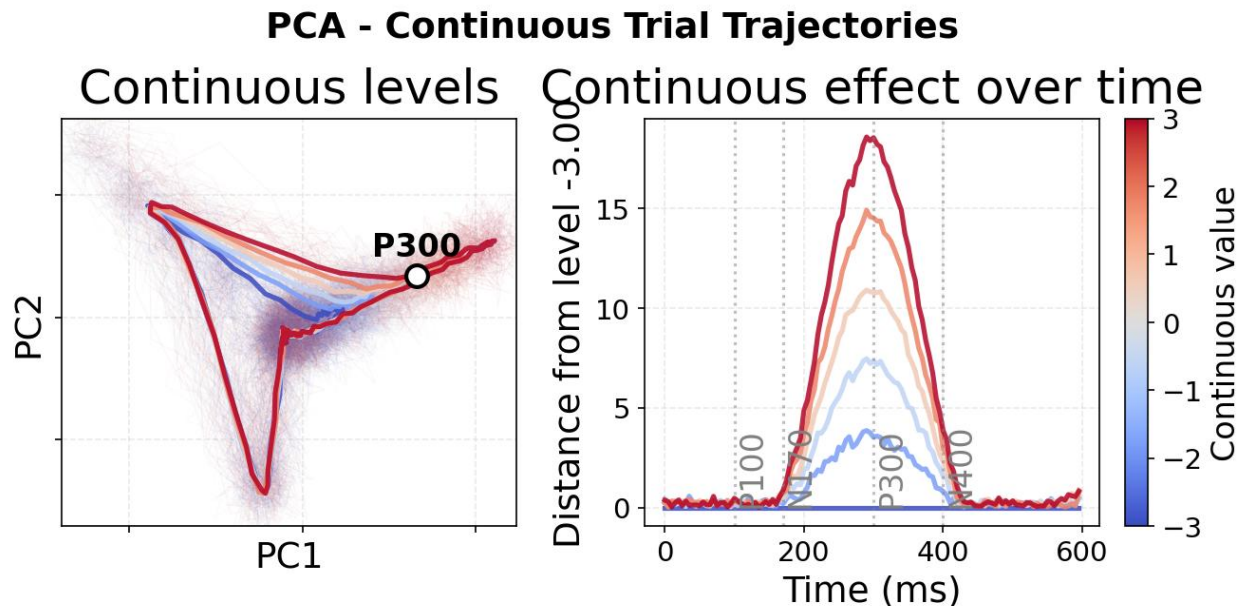
PCA - Individual Trial Trajectories (PC1=64.4%, PC2=22.0%)

Grand average (200 trials shown)



- PCA finds the 2 directions in channel space that explain the most variance
- Project all individual trials onto these same axes

PCA baseline comparison



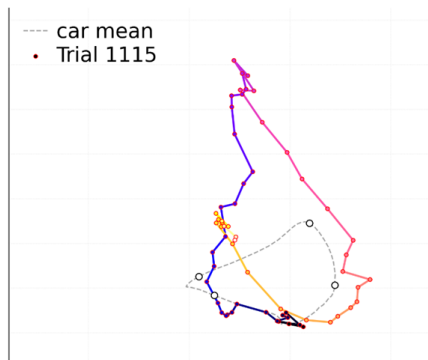
Angular and less smooth trajectories due to linear nature of PCA

Use case: Outlier detection

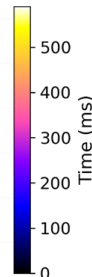
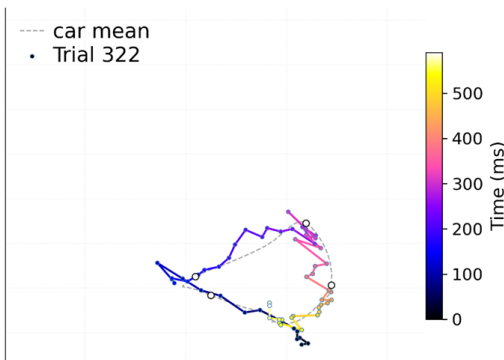
Approach 2

Outlier detection

Worst outlier
score 3.335



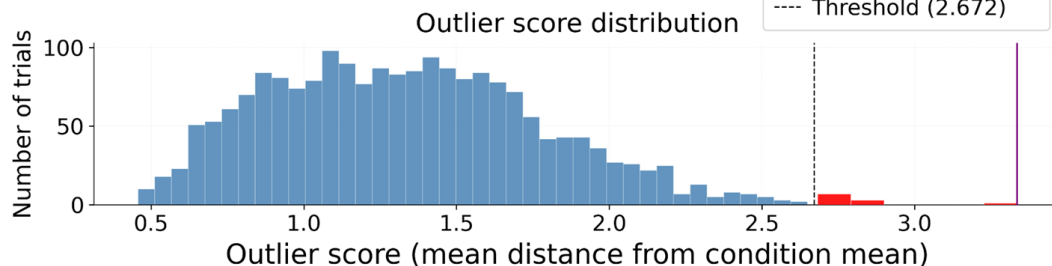
Typical trace
score 0.457



The outlier score is calculated by:

Deviation of each trial from its condition's mean trajectory in BCNE embedding space

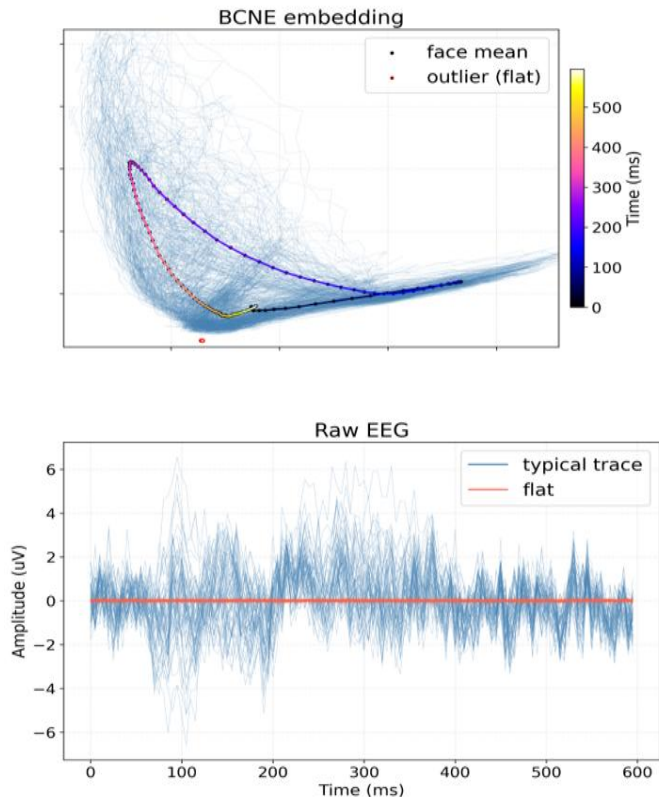
Threshold: > 3 times standard deviation



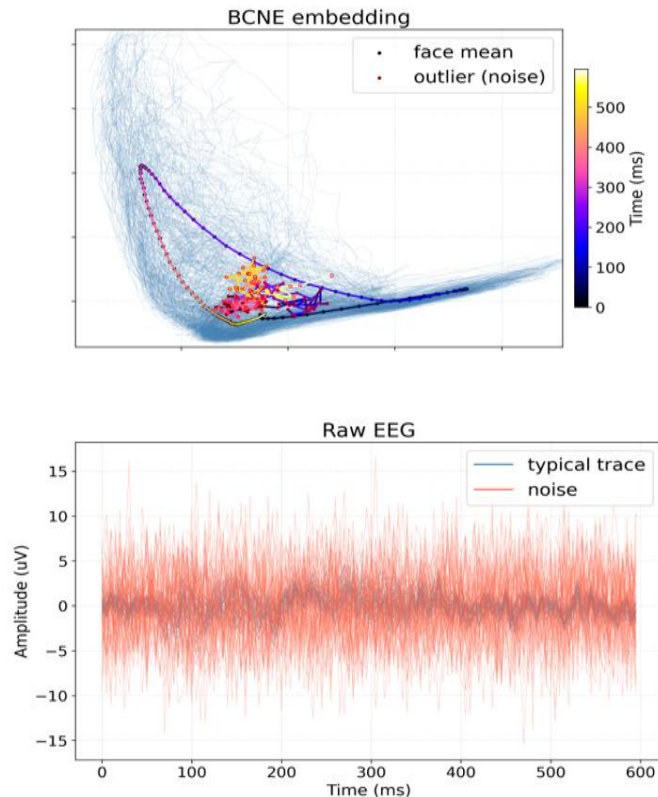
Flat and noisy trial projections

Approach 2

Outlier: 'flat' for face condition
Blue=typical trace trials, Red=outlier



Outlier: 'noise' for face condition
Blue=typical trace trials, Red=outlier



Discussion

Limitations

- T-PHATE and BCNE are designed for continuous data: applying to epoch-based EEG requires trial averaging, discarding trial-level information as seen in Approach 1
- Trial-level projection not evaluated for UMAP, PHATE, and T-PHATE
- No dedicated metric for temporal structure preservation
- Validated on simulated data only

Future Scope

- Validate on real EEG data
- Develop a dedicated temporal structure preservation metric
- Extend trial-level projection to other methods (such as PHATE, T-PHATE, and UMAP)

Conclusion

RQ1: Time-aware vs. Time-agnostic Methods

- BCNE and T-PHATE preserved temporal structure
- Time-agnostic methods produced fragmented ERP trajectories

RQ2: Approach 1 - Average by Condition

- Qualitative metrics were informative; no dedicated quantitative metric exists

RQ2: Approach 2 - Trial-level Projection

- BCNE fixed coordinate system enables trial projection without retraining
- Condition separation preserved at trial level
- Outlier detection enabled at trial level

References

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Thank you 😊

Questions?